

ENPM 645 – HUMAN ROBOT INTERACTION

FINAL PROJECT REPORT – [Varun Lakshmanan – 120169595]

Group - 7

Gesture-Based Control in Assistive Technology

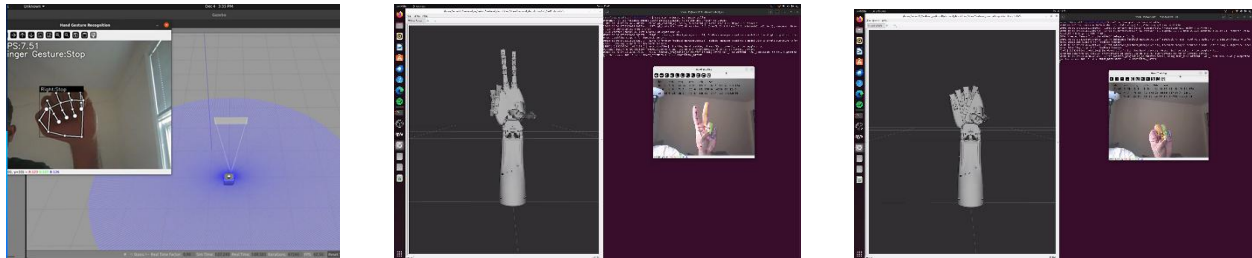
Project Summary:

The Project aimed to create a user-friendly interface for controlling assistive devices using hand gestures. This initiative was driven by the growing demand for assistive technologies in home and healthcare environments, particularly individuals with mobility challenges. This Project demonstrates two major applications:

1. **Controlling TurtleBot3 using Hand Gestures:** Implemented ROS2 node to control a Turtlebot3 using classified hand gestures detected using machine learning algorithm.
2. **Controlling a Robot Hand in Simulation:** Implemented ROS2 node to control a Dex Hand in Rviz using real-time human hand movements captured through webcam.

These applications highlight the system's versatility and potential to revolutionize human-robot interaction (HRI). By bridging the gap between human hand gestures and robotics actions, the project highlights how gesture – based control can be adapted for various real-world scenarios, from assistive technologies to industrial and research applications.

Results Achieved:



The above images show the outcome of the project. The First image shows the **user controlling a TurtleBot3 through Tele-Operation using hand gestures in Gazebo Environment**. These hand gestures are captured through a webcam which is further processed using Mediapipe Library, that detects the hand landmarks of hand and extracts key features such as finger and joint positions. These extracted features are further fed into the machine learning model that classifies the gestures. The recognized gestures are shown on the GUI -display. We can also see the robot stopping to the gesture signaling “STOP”.

The Second and Third images show our **implementation of DexHand hand control in simulation, using real-time detection of hand movement** captured as video feed through the webcam using Mediapipe libraries. Here, the Mediapipe processes 21 key landmarks and detects the joint and finger positions of the wrist, knuckles and fingertips, the data captured is further subjected to normalization for the robot hand to perform consistently regardless of the size of the hand. The system calculates joint angles for 24 degrees of freedom, including the wrist and individual fingers. To reduce noise caused by the Mediapipe and ensure a stable output, a deadband filter is applied to ignore unwanted minor movements and we also applied smoothing techniques to refine the data. These Joint angles are then published to ROS2 topic called `/joint_states`, allowing a simulation of DexHand to mimic the user's gesture in real time. The images show

the user controlling the DexHand which is visualized using Rviz, a 3D visualization tool. We can see the DexHand mirroring exact poses shown in the GUI like peace sign and clenched fist.

Success Metrics: Achieved a response time of 137ms from gesture detection to command execution and achieved 93% accuracy in detecting the predefined hand gestures using Mediapipe. Successfully eradicated the flickering issue caused by the noises from the data by normalizing it. Achieved 90% performance of the hand simulation (except for a few discrepancies in thumb movements).

Video Simulations:

[Turtlebot Simulation Using Hand Gestures](#) , [DexHand Control in ROS2](#)

Individual Contribution:

My Contributions to this Project focused on the DexHand Control and the required Mediapipe implementation for detection for joint positions, which is further used for joint calculations:

1. **Integration of Mediapipe with ROS2:** I Created a custom Handcontroller package in ROS2 and implemented Mediapipe for detecting hand landmarks from the webcam feed. This enabled real-time tracking of the fingers and joint positions for further processing.
2. **Data Normalization:** I implemented data normalization techniques for the joint positions captured by the Mediapipe. This reduced flickering issues that caused the robot hand to reset to its origin pos, frequently.
3. **Joint Angle Calculation:** I Calculated Joint angles for the DexHand, including the wrist, fingers and thumb, based on the detected movements of the Human Hand. These angles are then published to the ROS2 topic for real time simulation.
4. **Deadband and Smoothing:** Implemented Deadband filters and Smoothing techniques for the calculated joint angles to improve stability.

The Knowledge of ROS2, Open CV, and Python helped me in completing these contributions, ensuring proper movement of DexHand in the simulation Environment.

Peer Review:

Group 10 – Their idea of implementing a GUI for controlling multiple robots was novel and innovative. The implementation of visualization of multiple robots in a single Rviz window and their idea of controlling them through voice-control in future sounds fascinating.

Group 11 – This team presented an innovative approach by integrating EEG and eye-tracking data to enhance emotion recognition, achieving a notable accuracy of 73.59%. The use of Multimodal fusion demonstrates strong potential for applications in a personalized therapeutic setting. This Project was something new to me and iam waiting to see future results.

Group 14 – Their idea of using VR based interface for Path planning of UR-10 robot was innovative and offers a unique approach to human-robot interaction. They presented the idea in a more interesting way stating its applications in the real-world scenario by showing the details of Market analysis.

Group 5 - The Medi Body project is creative and provides practical solutions for healthcare automation, addressing important needs like medication delivery, patient monitoring, and caregiver support. I appreciate how seamlessly Natural Language Communication, AI and Robotics are integrated to enhance the efficiency in dynamic healthcare environments. I think the future results of this project might give a clear idea about it and iam eager to know about how it works.